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Module 6:

Magnetic quantities

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Inhaltsverzeichnis

1 Introduction

Electromagnetism forms the basis of many technologies in our everyday lives. For example, magnetic resonance imaging enables precise diagnoses and therapies in medicine. Generators and transformers ensure the efficient generation, transmission and distribution of electrical energy. In computer technology, magnetic storage devices such as hard drives are indispensable for data processing and storage. However, the limits of their application areas have not yet been reached. Future technological developments such as quantum computers, wireless power supply and magnetic levitation will push these limits even further. A sound understanding of magnetic quantities is crucial for the further development and optimisation of existing and future technologies.

This chapter provides an introduction to electromagnetic effects and the associated magnetic quantities. First, the magnetic field is described, followed by an explanation of relevant quantities for the magnetic circuit, such as flux, magnetic flux and magnetic resistance. It then explains how the Lorentz force, induction and inductance work, followed by a calculation of the energy content in the magnetic field. The chapter concludes with a digression on the skin effect and Hall effect.

Learning objectives: Magnetic quantities

The Students

- know the basic magnetic quantities in the magnetic circuit.
- understand the physical principles behind the individual magnetic quantities.
- can describe the interactions between the magnetic quantities.
- can calculate the individual quantities in the magnetic circuit.

1.1 Magnetism

Magnetism is a physical phenomenon that manifests itself in the form of interacting forces between magnetised or magnetisable solids and moving electric charges. These forces are represented by a magnetic field. The most common forms of magnetism are electromagnetism and the magnetism of solids. Ferromagnetism is the best known and most important type of magnetised solid and describes the magnetic behaviour of some metallic bodies, known as ferromagnetic materials. As can be seen in Figure 1.1, a ferromagnetic material such as iron, cobalt or nickel consists of many small elementary magnets in so-called Weiss domains, which are separated from each other by Bloch walls. In a non-magnetised body, these Weiss domains are arranged randomly and are therefore not aligned. The body is not (or only slightly) magnetised on the outside, as the magnetisation directions of the individual domains largely cancel each other out. Durch ein starkes externes Magnetfeld (die benötigte Stärke ist temperatur- und materialabhängig) können die ungeordneten Elementarmagnete parallel ausgerichtet werden. Durch die gleiche Ausrichtung der Weissschen Bezirke wird das Material selbst magnetisch.

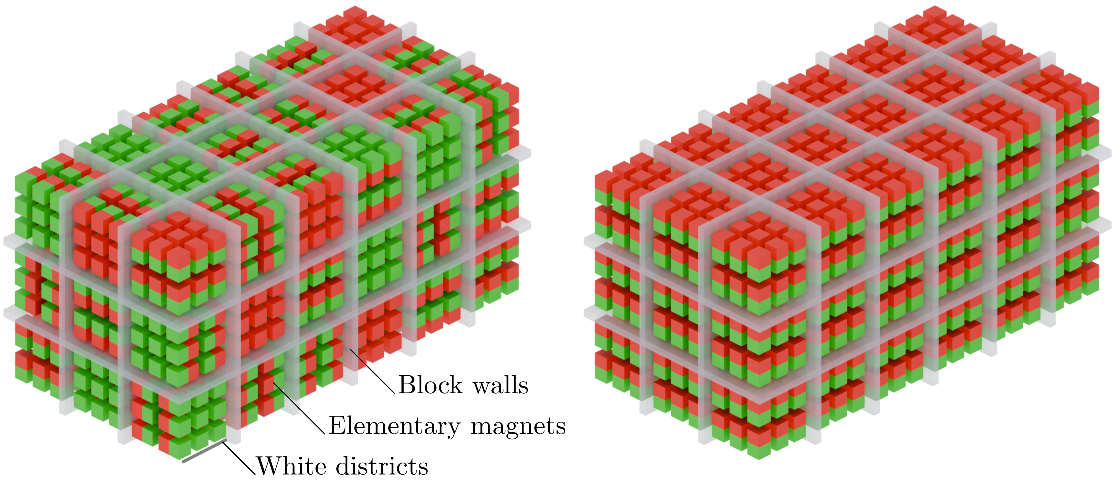

Figure 1.1: **Structure of a permanent magnet.** On the left, the Weiss domains with the elementary magnets are randomly aligned, and the body is not magnetised. The body on the right is magnetised because the elementary magnets of the Weiss domains are aligned in the same order.

1.2 Magnetic field

The orderly alignment of the Weiss domains creates a magnetic field that can be measured outside the body. Magnetic fields are vector fields that exert a force on magnetic materials in space. The strength of the magnetic field is described by the magnitude of the magnetic field strength $\vec{H}$. This vector quantity assigns a corresponding direction with a specific magnetic strength to each point in space on which the magnetic field acts. Magnetic fields and the associated magnetic field strength are represented by field lines. Magnetic field lines have characteristic properties:

- Magnetic field lines are always closed (source-free).
- Outside the magnet, they run from the north pole to the south pole.
- They always exit or enter the magnetic surface perpendicularly.

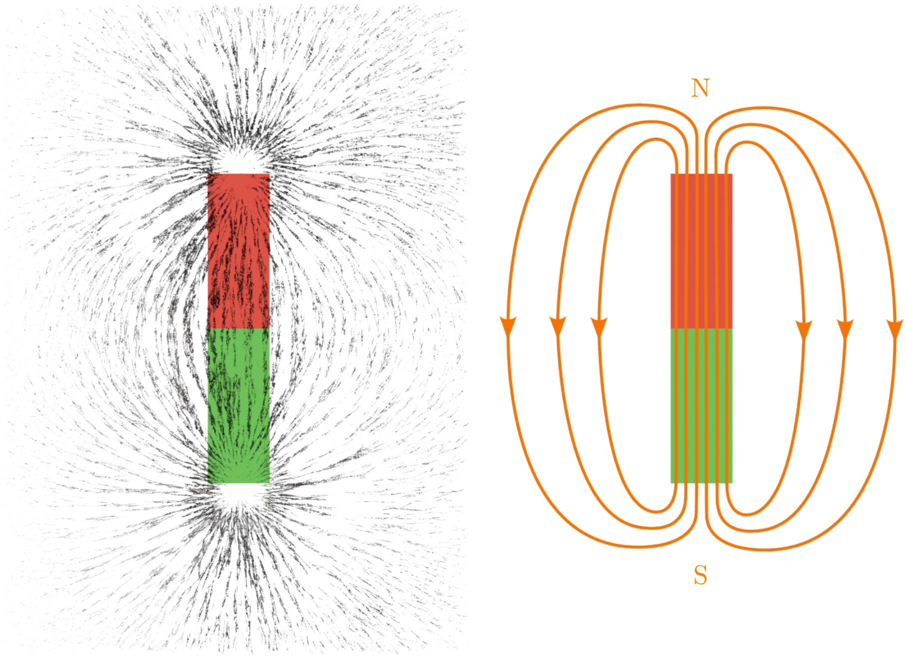

Figure 1.2: **Field lines of a permanent magnet.** On the left, the field lines were visualised using metal filings above a magnet. The graphic on the right shows the properties of magnetic field lines: closed, running from north to south outside the magnet, exiting the magnet perpendicularly.

Key point: Magnetic field

Magnetic fields are vector fields that have a specific direction and strength at each point in space where they exert their influence. The strength and direction are described by the magnetic field strength $\vec{H}$.

2 Electromagnetism

A magnetic field can be generated not only by a permanent magnet, but also by an electric current. This results in a circular magnetic field around the current-carrying conductor. The direction of the field can be determined using the „right-hand rule“: When the hand is formed into a fist with the thumb extended, the thumb points in the direction of the (technical) electric current flow (the „technical current flow“runs from the positive to the negative pole) and the fingers point in the direction of the magnetic field lines.

Key point: Right hand rule

If the thumb of the right hand points in the direction of the (technical) electric current flow, the remaining bent fingers indicate the direction of the magnetic field lines.

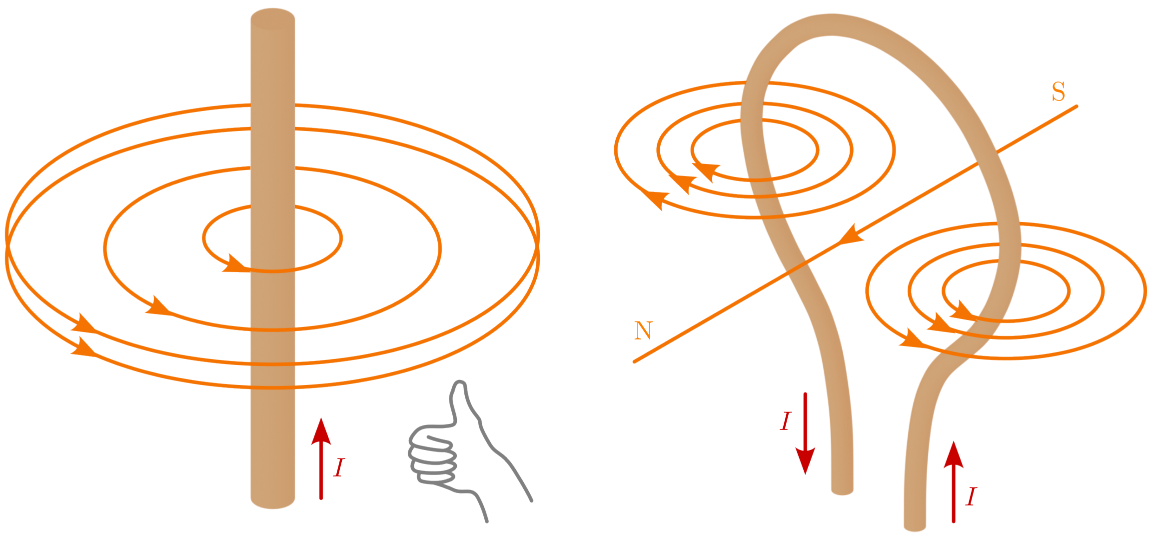

Figure 2.1: **Magnetic field around a current-carrying conductor.** Left: The direction of the field can be illustrated using the right-hand rule. Right: The conductor loop represents the smallest unit of a coil. The middle field line shown here closes at infinity.

2.1 Magnetic flux

The magnetic fields generated by electric currents are measured using the parameter magnetic flux density Θ (theta). Analogous to electric voltage, flux density is also referred to as magnetic voltage. Since flux density results from current, it has the same unit as current, namely amperes.

The name is derived from the flux law. This states that the magnetic flux Θ is equal to the total current I of an area through which it flows (see Figure 2.2). Equation 2.1 represents the flux law in its general form. The right-hand side denotes the surface integral of the current density $\vec{J}$. This integral expresses the sum of the current flowing through the surface A. The left-hand side of the equation represents the closed line integral over the magnetic field strength $\vec{H}$. This closed line $\mathrm{d}\vec{s}$ corresponds to the boundary of the surface A.

$$\Theta = \oint_s \vec{H} \cdot \mathrm{d}\vec{s} = \iint_A \vec{J}\,\mathrm{d}\vec{A} \tag{2.1}$$

Since in most cases the current is transported through a conductor and consequently both the direction and the strength of the current are clearly known, it is sufficient in this case to replace the integral (as shown in equation 2.2) with the number of current-carrying conductors N multiplied by the current strength I.

$$\Theta = \oint_s \vec{H} \cdot \mathrm{d}\vec{s} = N \cdot I \qquad [\mathrm{A}] \tag{2.2}$$

Key point: Magnetische Durchflutung

The magnetic flux Θ corresponds to the total current of an area through which it flows.

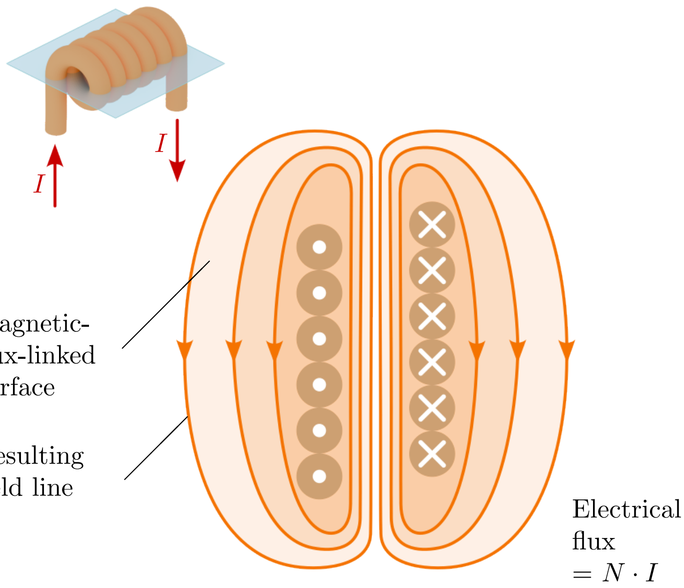

Abbildung 2.2: **Flushing of a coil.** The flux corresponds to the total current through a fluxed area. The electrical flux of a coil therefore depends on the number of windings and the current strength. For representation in a two-dimensional graphic, the symbol ⊗ is used for the current flow „into the image area“, ⊙ for „out of the image area“.

If the magnetic field strength is constant over the integration path $\mathrm{d}\vec{s}$, the vector $\vec{H}$ can be moved outside the integral. This condition is usually fulfilled if the field line is located in the same material over the entire integration path. An example would be a circular field around a conductor at the centre of the circle or a toroidal coil, as shown in Figure 2.3. In this case, the integral $\oint \vec{H} \cdot \mathrm{d}\vec{s}$ becomes the length of the integration path (Equation 2.3). This length is called the mean field line length and is denoted by ℓ_m. It represents the mean value of the sum of all field lines located within the circular field. The magnetic field strength in such arrangements can be easily determined using equation 2.4.

$$\Theta = \oint_s \vec{H} \cdot \mathrm{d}\vec{s} = |\vec{H}| \cdot \ell_\mathrm{m} \tag{2.3}$$

$$|\vec{H}| = \frac{\Theta}{\ell_\mathrm{m}} \qquad \left[\frac{\mathrm{A}}{\mathrm{m}}\right] \tag{2.4}$$

Key point: Calculation aid for magnetic field strength

If the magnetic field strength is constant over the entire path, it can be calculated by the quotient of flux Θ and path length ℓ_m. Its unit is ampere per metre.

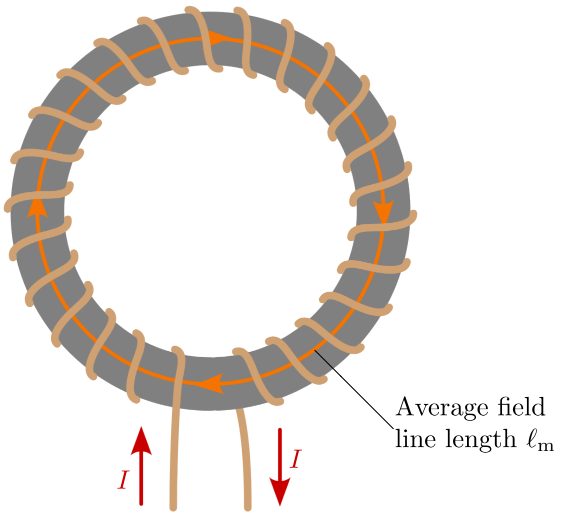

Figure 2.3: **Toroidal coil with medium field line length.** The average field line position represents the mean value of all field lines within the coil, which means that the field strength can be easily calculated.

Example 2.1: Magnetic field strength over the mean field line length

A toroidal coil (Figure 2.3) with 1000 turns and an average field line length of 50 cm is traversed by a current of 100 mA. How strong is the magnetic field?

$$\varTheta = H \cdot \ell_\mathrm{m} = N \cdot I$$
$$H = \frac{N \cdot I}{\ell_\mathrm{m}} = \frac{1000 \cdot 0{,}1\,\mathrm{A}}{0{,}5\,\mathrm{m}} = 200\,\frac{\mathrm{A}}{\mathrm{m}}$$

Example 2.2: Magnetic field strength across the circumference

A direct current of $I = 50\,\mathrm{A}$ flows through a straight conductor. How strong is the magnetic field at a distance of $r = 20\,\mathrm{cm}$?

$$H = \frac{N \cdot I}{\ell_\mathrm{m}} = \frac{1 \cdot 50\,\mathrm{A}}{2\pi \cdot 0{,}2\,\mathrm{m}} = 39{,}79\,\frac{\mathrm{A}}{\mathrm{m}}$$

The mean field line length ℓ_m can be determined by the circumference of the circle and is therefore replaced by the formula for calculating the circumference of a circle, $2\pi \cdot r$.

2.2 Magnetic flux and flux density

Magnetic flux is comparable to electric current in an electric circuit. Contrary to the terminology, however, there is no flow of magnetic particles; rather, it is symbolically understood as „the amount of magnetic field“ and acts as a result of magnetic tension. The symbol for magnetic flux is $\varPhi$, and the unit is the weber (Wb). One weber (Wb) is equivalent to one volt-second (Vs).

Apart from the magnetic flux, the force exerted by a magnet also depends on the area through which the flux passes. The more concentrated the field lines are, the greater the magnetic effect.

This is described by the magnetic flux density B, which in the simplest case (non-curved surface, homogeneous flux density) is defined by the quotient of the magnetic flux Φ and the area A. The unit of flux density is Tesla (T). The direction of the flux density $\vec{B}$ is perpendicular to the area, which is expressed by the normal vector to the area $\vec{A}$.

$$\vec{B} = \frac{\Phi}{\vec{A}} \qquad [\mathrm{T}] \tag{2.5}$$

In general (without the above restrictions), the following applies:

$$\Phi = \iint_A \vec{B} \cdot \mathrm{d}\vec{A} \tag{2.6}$$

Key point: Magnetic flux density

The magnetic flux density $\vec{B}$ describes the concentration of the magnetic flux Φ perpendicular to a surface A.

Both magnetic field strength and magnetic flux density are vector quantities. They can therefore be represented graphically by field lines. Magnetic flux Φ, on the other hand, is a scalar quantity.

The magnetic flux density $\vec{B}$ and the magnetic field strength $\vec{H}$ are linked via the permeability μ. Permeability consists of the product of a material-independent parameter, the magnetic field constant $\mu_0 = 1.256\,637\,062 \cdot 10^{-6}\,\frac{\mathrm{Vs}}{\mathrm{Am}}$, and a material-specific permeability μ_r. The magnetic field constant describes the permeability in a vacuum and, until the reorganisation of the SI units in 2019, was precisely defined with the value $\mu_0 = 4\pi \cdot 10^{-7}\,\frac{\mathrm{Vs}}{\mathrm{Am}}$. It is now subject to measurement uncertainty.

Table 2.1: **Material-dependent permeability.** Exemplary overview of the permeability coefficient μ_r of different materials:

Material	**Permeability coefficient** μ_r
Water	$1 - 9{,}1 \cdot 10^{-6}$
Copper	$1 - 6{,}4 \cdot 10^{-6}$
Air	$1 + 4 \cdot 10^{-7}$
Aluminium	$1 - 2{,}2 \cdot 10^{-5}$
Iron	300 to 140000

$$\vec{B} = \mu_0 \cdot \mu_r \cdot \vec{H} \tag{2.7}$$

In a ferromagnetic material, the relationship between the magnetic field strength $\vec{H}$ and the magnetic flux density $\vec{B}$ is not linear. The permeability reaches saturation as the magnetisation increases, so that the relative permeability μ_r approaches 1 from a material-dependent initial value. If the magnetic field is reduced again (or set to zero), the magnetisation remains to a certain extent (point B_r in Figure **??**). This process is called remanence. The remaining magnetic flux density at a magnetic field strength of zero is the remanence flux density B_r. To completely demagnetise the material again, an inverse magnetic field strength, the coercive field strength H_c, is required.

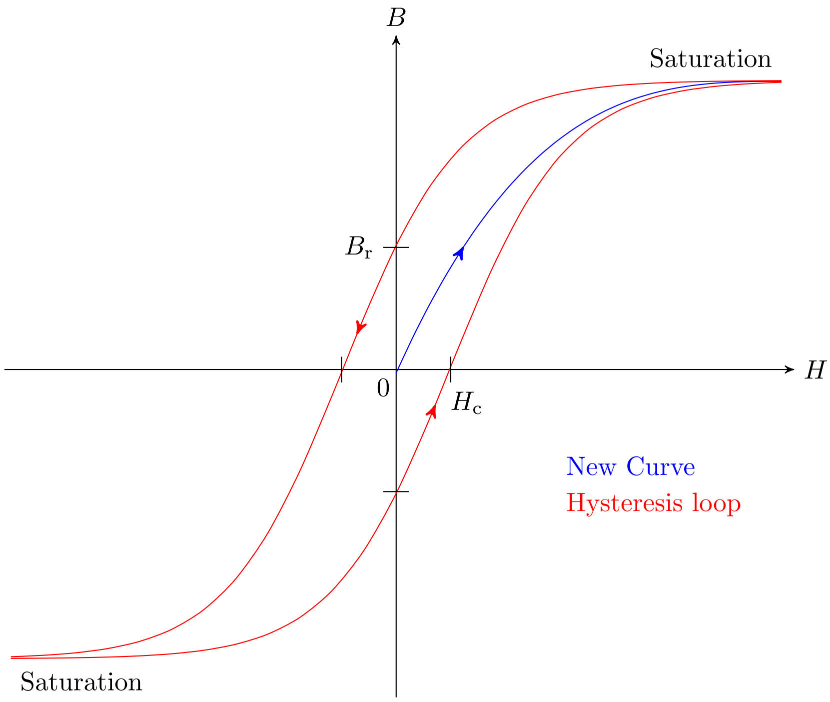

Abbildung 2.4: **Magnetisation curve of a ferromagnetic material.** The magnetic field strength and the magnetic flux density do not behave linearly in relation to each other.

Ferromagnetic materials are divided into hard and soft magnetic materials based on their coercive field strength. Hard magnetic materials (e.g. strong permanent magnets made of neodymium- iron-boron) have a value for H_c greater than $10 \cdot 10^3 \ frac{\mathrm{A}}{\mathrm{m}}$, while for soft magnetic materials (e.g. magnetic cores made of manganese-zinc ferrite), H_c is less than $500\,\frac{\mathrm{A}}{\mathrm{m}}$. Hard magnetic materials are mainly used for permanent magnets.

Example 2.3: Magnetic flux density

Inside a tightly wound toroidal coil, a magnetic field strength of $H = 100\,\frac{\mathrm{A}}{\mathrm{m}}$ is to be generated. The coil has a mean radius of 5 cm.

1. Calculate the required current I if the coil has $N = 200$ windings.

From equation 2.2 and 2.3:

$$\Theta = H \cdot \ell_\mathrm{m} = N \cdot I$$
$$I = \frac{H \cdot \ell_\mathrm{m}}{N} = \frac{100\,\frac{\mathrm{A}}{\mathrm{m}} \cdot 2 \cdot \pi \cdot 5 \cdot 10^{-2}\,\mathrm{m}}{200} = 157{,}08\,\mathrm{mA}$$

2. How large will the flux density B be in the case of an air coil ($\mu_\mathrm{r} = 1$) or an iron-filled coil ($\mu_\mathrm{r} = 2000$ at the operating point)?

From equation 3.3:

$$B_\mathrm{Luft} = \mu_0 \cdot \mu_\mathrm{r} \cdot H = 1{,}256 \cdot 10^{-6}\,\frac{\mathrm{Vs}}{\mathrm{Am}} \cdot 100\,\frac{\mathrm{A}}{\mathrm{m}} = 125{,}6\,\mu\mathrm{T}$$
$$B_\mathrm{Eisen} = 2000 \cdot B_\mathrm{Luft} = 251{,}2\,\mathrm{mT}$$

3 Magnetic flux and flux density

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$$\vec{B} = \frac{\Phi}{\vec{A}} \qquad [\mathrm{T}] \tag{3.1}$$

In general (without the above restrictions), the following applies:

$$\Phi = \iint\limits_{A} \vec{B} \cdot \mathrm{d}\vec{A} \tag{3.2}$$

Key point: Magnetic flux density

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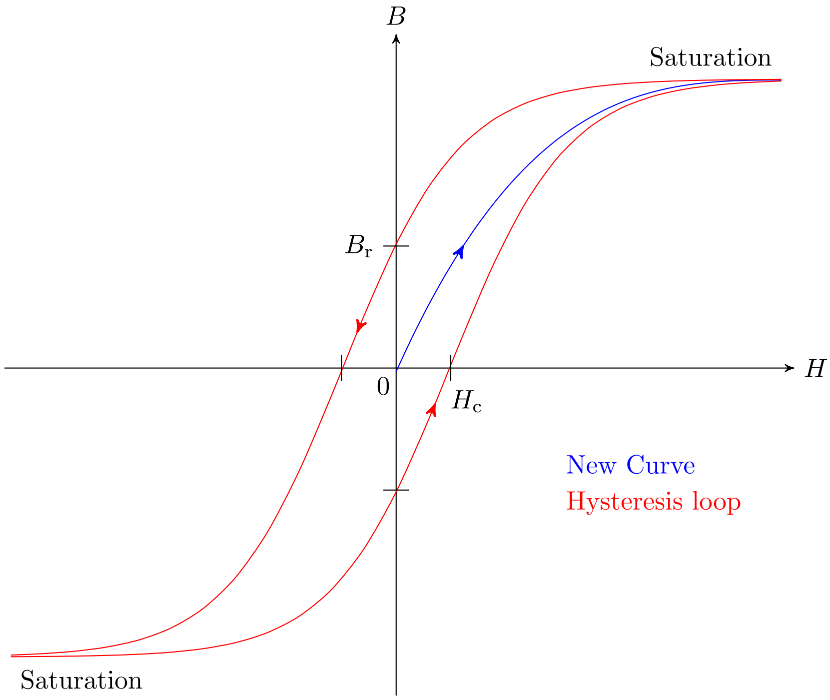

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$$B_\mathrm{Eisen} = 2000 \cdot B_\mathrm{Luft} = 251{,}2\,\mathrm{mT}$$

4 Magnetic resistance

Magnetic tension, also known as flux, was already discussed in chapter 2.1, and magnetic flux, which is the equivalent of electric current, was discussed in chapter 3. It therefore stands to reason that, in a magnetic circuit, there is also magnetic resistance, which is equivalent to ohmic resistance. It has the formula symbol R_m and the unit $\frac{\mathrm{A}}{\mathrm{Vs}}$ and is also called reluctance .

An example of a magnetic circuit is shown in Figure **??** by a simple iron ring with a single-sided coil. The coil generates a magnetic flux Θ analogous to electrical voltage. The iron circuit consists of four partial resistances, since the magnetic resistance depends on both the length and the cross-section through which the current flows. The four resistances are traversed by the magnetic flux Φ.

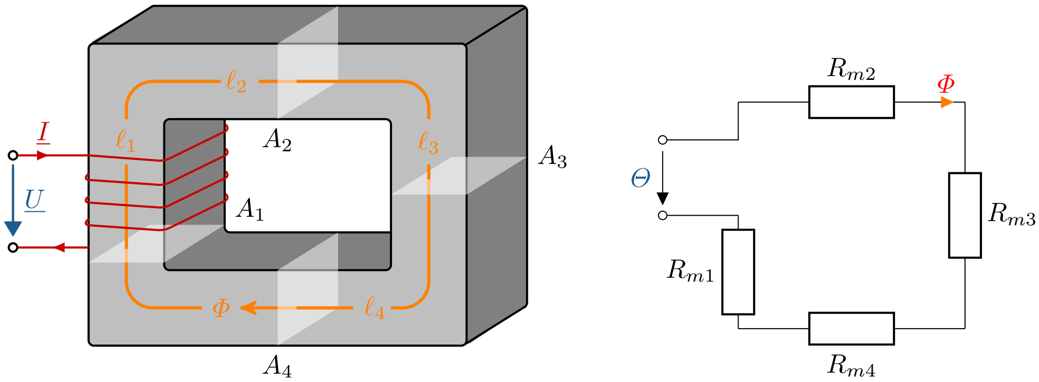


Figure 4.1: **Magnetic circuit with an iron ring.** The four magnetic partial resistances depend on the length, material and cross-section of the iron ring through which the current flows.

In simple arrangements (as shown in Figure **??**, for example), the resistance can be expressed by equation 4.1, neglecting the corners. As with the magnetic field strength, ℓ_m is the average field line length of the resistance within the magnetic circle. To determine the average field line length ℓ_m, the lengths of all sides are added together. A is the cross-sectional area through which the magnetic flux flows. μ is the permeability of the material. The magnetic resistance R_m is now calculated from the sum of the mean field line lengths ℓ_m divided by the product of the cross-sectional area A through which the magnetic flux flows and the permeability $\mu_\mathrm{r} \cdot \mu_0$.

$$R_\mathrm{m} = \frac{\ell_\mathrm{m}}{\mu_\mathrm{r} \cdot \mu_0 \cdot A} \qquad \left[\frac{\mathrm{A}}{\mathrm{V} \cdot \mathrm{s}}\right] \tag{4.1}$$

Key point: Magnetic resistance

To calculate the magnetic resistance R_m, the average field line length ℓ_m is divided by the product of the permeability of the material $\mu_\mathrm{r} \cdot \mu_0$ and the cross-sectional area A. If the material properties or cross-sectional areas within the magnetised body vary, the partial resistances are first calculated and then added together to give the total resistance.

4.1 The magnetic circle

If we now consider all known magnetic quantities in their interaction, we obtain the magnetic analogue of Ohm's law. It describes that the magnetic voltage Θ corresponds to the product of the magnetic resistance R_m and the magnetic flux Φ.

$$\Theta = R_\mathrm{m} \cdot \Phi \tag{4.2}$$

Key point: Relationships between magnetic field magnitudes

The magnetic field strength H in the special case of a toroidal coil is calculated from the product of the number of turns N and the current flow I, divided by the mean field line length ℓ_m.

$$H = \frac{N \cdot I}{\ell_\mathrm{m}}$$

The magnetic flux density B is determined by multiplying the magnetic field strength H by the permeability $\mu_\mathrm{r} \cdot \mu_0$.

$$B = \mu_\mathrm{r} \cdot \mu_0 \cdot H$$

The magnetic flux Φ is obtained from the integral of the magnetic flux density B over an area A.

$$\Phi = \iint_A \vec{B} \cdot \mathrm{d}\vec{A}$$

5 Lorentz force

If a conductor carrying current is placed in a magnetic field, a force acts on it that is perpendicular to the conductor and the magnetic field. This force is called the Lorentz force. If the conductor is perpendicular to the magnetic field lines, the force is greatest.

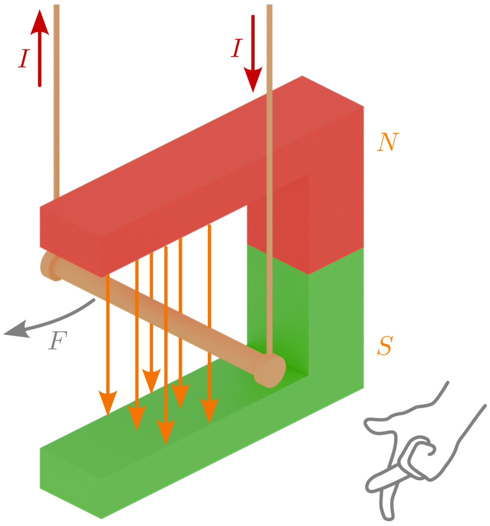
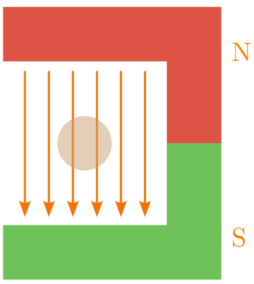


Figure 5.1: **Experimental setup for the Lorentz force.** Deflection of a freely oscillating conductor carrying current in the field of a permanent magnet, causing the conductor to move in the direction of the force. The direction can be determined using the three-finger rule of the right hand. The thumb points in the direction of the current flow, the index finger in the direction of the magnetic field lines. The middle finger then points in the direction in which the force acts.

Step-by-step explanation of the experimental setup shown in Figure 5.1 with a current-carrying conductor in the field of a permanent magnet:



- A conductor is located between the poles of a horseshoe magnet.
- The conductor is not electrically conductive here, so that the field of the permanent magnet can be considered individually.
- The magnetic field lines run from the north pole to the south pole outside the permanent magnet and close inside the magnet.

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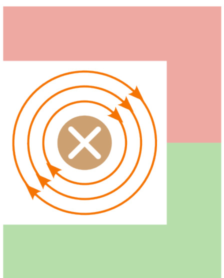

- Now the conductor is considered to be carrying current, without taking the field of the permanent magnet into account.
- A magnetic field is created around the current-carrying conductor depending on the direction of the current flow (right-hand rule).

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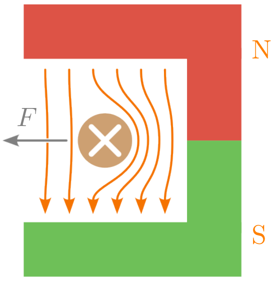
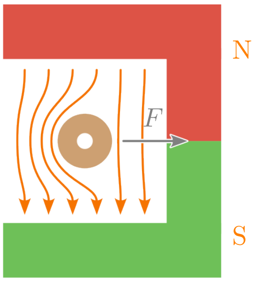


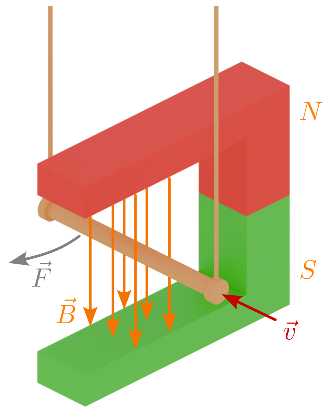
- The field of the permanent magnet and the magnetic field of the current-carrying conductor are now considered together.
- Both fields overlap, resulting in a field amplification on the right side and a field attenuation on the left side.
- The system attempts to compensate for this field distortion, causing a force to act on the conductor that is perpendicular to the conductor and the magnetic field (three-finger rule).
- The conductor moves in the direction in which the force acts.



- Here, the direction of flow of the current-carrying conductor is reversed.
- The Lorentz force acts in the opposite direction (three-finger rule).
- The conductor moves in the direction in which the force acts.

The vector or scalar representation can be used to calculate the Lorentz force. In the vector calculation (equation 5.1), a moving unit charge q with velocity $\vec{v}$ in the magnetic field $\vec{B}$ causes the force $\vec{F}$:

$$\vec{F} = q \cdot (\vec{v} \cdot \vec{B}) \tag{5.1}$$



In the scalar calculation, the vectors are omitted and instead the angle sin α in relation to the velocity v and the magnetic field B are included in the calculation. It applies that the Lorentz force is perpendicular to both the flux density and the velocity of the charge. The magnitude of the force is calculated for straight conductors and a constant magnetic field as follows:

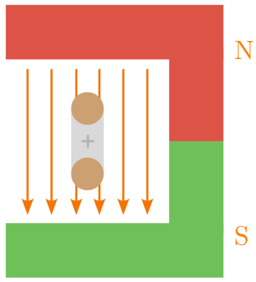
$$F = q \cdot v \cdot B \cdot \sin\alpha \tag{5.2}$$
$$F = I \cdot \ell \cdot N \cdot B \cdot \sin\alpha \tag{5.3}$$

Equation 5.3 illustrates the scalar calculation in a current-carrying conductor. ℓ is the effective conductor length with the number of N parallel conductors carrying the current I. α is the angle between the conductor and the direction of the magnetic flux density $\vec{B}$. If the conductor and the magnetic field are perpendicular to each other, $\sin\alpha$ is omitted.

Key point: Lorentz force

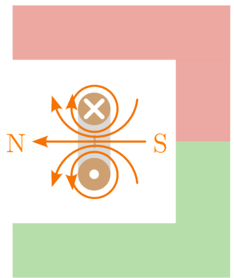
The Lorentz force describes the force acting on a moving charge in a magnetic field. If the charge flows through a conductor, the Lorentz force acts perpendicular to both the current flow and the magnetic flux density.

To illustrate the principle of an electric motor, consider a current-carrying conductor loop that is movably attached to a rotary rotor in the field of a permanent magnet:



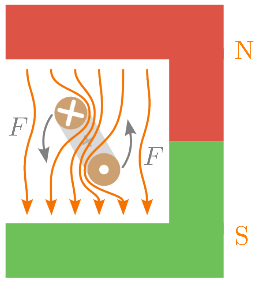
- A conductor loop is located on a rotating element, the rotor, in the magnetic field of a horseshoe magnet.
- No current is flowing through the conductor yet, so the field of the permanent magnet can be considered individually.
- The magnetic field lines run outside the permanent magnet from the north pole to the south pole.

+

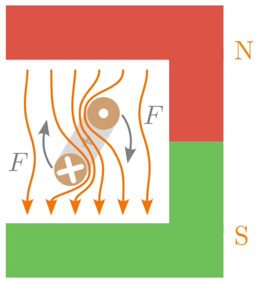


- Now consider the current flowing through the conductor loop without taking the field of the permanent magnet into account.
- A magnetic field is created around the current-carrying conductor loop depending on the direction of the current flow (right-hand rule).

⇓



- The field of the permanent magnet and the magnetic field of the current-carrying conductor loop are now considered together.
- The Lorentz force, which is perpendicular to the conductor and the magnetic field, acts on the conductor loop.
- The conductor loop begins to rotate due to the Lorentz force.



- Here, the direction of flow of the current-carrying conductor is reversed.
- Accordingly, the polarity of the current-carrying conductor loop and thus the direction of rotation of the conductor loop are also reversed (three-finger rule).

Example 5.1: Lorentzkraft

A direct current motor has a magnetic flux density of $B = 0.8\,\mathrm{T}$ in the air gap. There are a total of $N = 400$ windings under the poles, through which a current of $I = 10\,\mathrm{A}$ flows. The effective conductor length is $\ell = 150\,\mathrm{mm}$.

Calculate the force F at the circumference of the anchor.

$$\begin{aligned} F &= B \cdot I \cdot \ell \cdot N \\ &= 0{,}8\,\tfrac{\mathrm{Vs}}{\mathrm{m}^2} \cdot 10\,\mathrm{A} \cdot 0{,}15\,\mathrm{m} \cdot 400 \\ &= 480\,\frac{\mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s} \cdot \mathrm{A} \cdot \mathrm{m}}{\mathrm{s}^3 \cdot \mathrm{A} \cdot \mathrm{m}^2} = 480\,\mathrm{N} \end{aligned}$$

Two parallel conductors through which current flows also exert a Lorentz force on each other, as each conductor through which current flows creates a magnetic field around itself (see Figure 5.2). If the direction of current is the same in both conductors, an attractive force is created; if the direction of current is reversed, the force is repulsive.

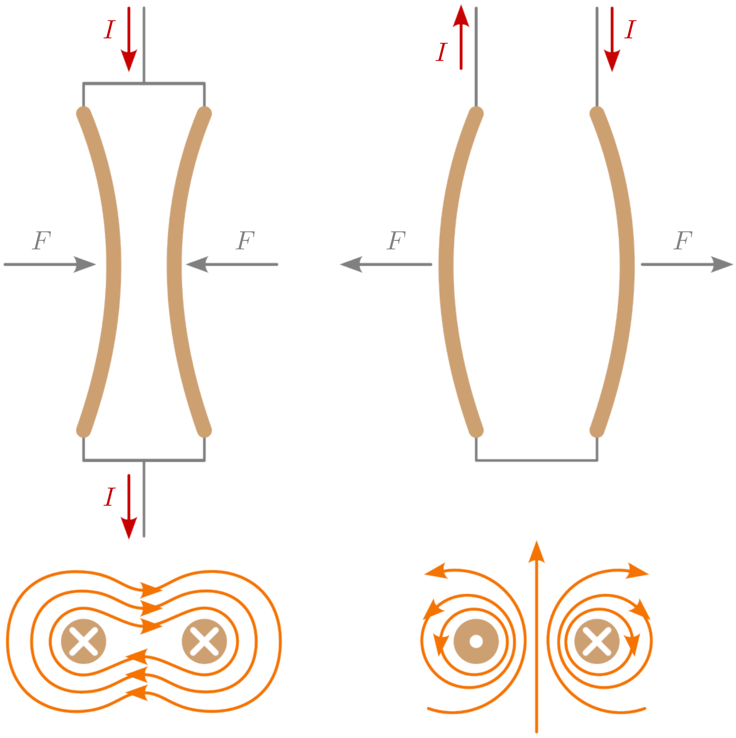


Figure 5.2: **Lorentz force between two parallel conductors.** If the direction of current flow is the same, they attract each other. If the direction of current flow is opposite, the conductors repel each other.

For the theoretical special case of two straight, parallel, thin and infinitely long wires, the following ratio applies:

$$F_{1,2} = \frac{\ell \cdot \mu_0 \cdot I_1 \cdot I_2}{2 \cdot \pi \cdot r} \tag{5.4}$$

Since these specific conditions do not apply in practice, the formula serves merely as an approximation of real cases in order to estimate the force exerted by the Lorentz force.

6 Induktion

Induction describes the generation of an electrical voltage in a conductor due to a change in magnetic flux over time. The change in magnetic flux can be caused in various ways. Figure 6.1 illustrates induction using a moving conductor in a magnetic field. As in the previous example (Figure 5.1), a conductor is penetrated by the magnetic field of a permanent magnet, but with the difference that this time no current flows through the conductor. Instead, the ends of the conductor are connected to a voltage measuring device. If the conductor is now moved by an external force, the Lorentz force acts on the electrons in the conductor. The movement causes a charge separation in the conductor. This charge separation is called induction voltage. As long as the conductor or the inducing magnetic field is in motion, a voltage can be read on the measuring device.

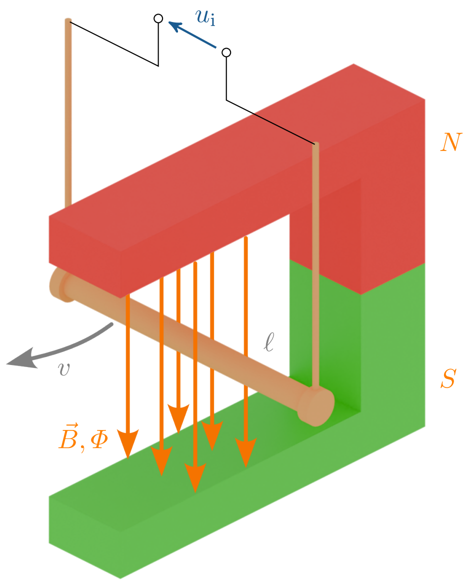


Figure 6.1: **Experimental setup for measuring an induced voltage.** If a time-varying magnetic field is generated by the movement of the conductor, the Lorentz force causes charge separation in the conductor, inducing a measurable voltage.

In simple terms, the induced voltage can be calculated using equation 6.1. The magnitude of the induced voltage u_i depends on the magnetic flux density B, the length of the conductor in the magnetic field ℓ, the speed of movement or flux change v, and the number of conductors in the magnetic field N. Since this voltage varies over time, the symbol u_i is written in lower case.

$$u_\mathrm{i} = B \cdot \ell \cdot v \cdot N \tag{6.1}$$

In general, the induced voltage u_i is the change in magnetic flux $\mathrm{d}\Phi/\mathrm{d}t$ over time multiplied by the number of conductors N. This is reflected in the general law of induction (equation 6.2). The negative sign is due to Lenz's law, which states that cause and effect always behave in opposite ways.

$$u_\mathrm{i} = -N \cdot \frac{\mathrm{d}\Phi}{\mathrm{d}t} \tag{6.2}$$

Key point: Induction

Induction is the electrical voltage u_i generated by a change in magnetic flux $\mathrm{d}\Phi/\mathrm{d}t$.

7 Induktivität

Inductance describes the ability to store electrical energy in a magnetic field. Components that have this ability are called coils (Figure 7.1), chokes or inductors.

Figure 7.1: **Air coil made of enameled wire.** Air coils do not have a magnetic core and therefore have a comparatively lower inductance than coils with a soft magnetic core.

The functioning of an inductor is based on the electromagnetic field generated by the flow of current. This electromagnetic field induces a current in adjacent conductor coils of the component. This effect allows electrical energy to be temporarily stored in a magnetic field. For a better understanding, the function of inductance is explained step by step below. A current flows through a conductor loop. This current generates a magnetic field whose strength is proportional to the current strength – the stronger the current, the stronger the magnetic field. If the current strength is continuously changed, this also causes a change in the magnetic field, which in turn (as described in chapter 6) induces a voltage in neighbouring conductor loops. This induced voltage counteracts the change in current according to Lenz's law. Consequently, when the current source is switched on at the coil, the induced voltage counteracts the current flow, so that the current only increases gradually. When the coil is switched off, the induced voltage causes the current to continue to flow for a while. This phenomenon, also known as self-induction, enables magnetic storage. The inductance therefore indicates how strongly an electrical component self-induces. The level of inductance L depends on the number of turns N, the permeability of the coil core μ_r, the cross-sectional area of the core and the geometric shape of the component. For inductors, the magnetic field is generally proportional to the value of the current.

The inductance L can be calculated in two different ways using the unit Henry (H). On the one hand, it can be calculated using the relationship between the resulting magnetic field $N \cdot \Phi$ and the causing current I (as in equation 7.1).

$$L = \frac{N \cdot \Phi}{I} \qquad [\mathrm{H}] \tag{7.1}$$

On the other hand, for simple geometric structures such as cylindrical, rectangular or toroidal coils, the inductance can be determined from the relationship between the number of turns N^2 and the magnetic resistance R_m (Equation 7.2).

$$L = \frac{N^2}{R_\mathrm{m}} \qquad [\mathrm{H}] \tag{7.2}$$

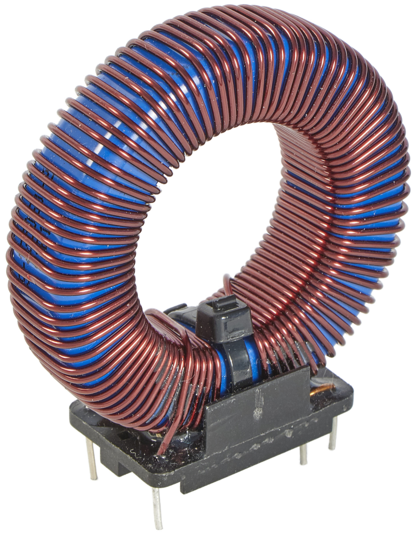


Figure 7.2: **Photograph of a toroidal coil.** The ring-shaped design has the special feature that the magnetic flux moves primarily within the core, thereby reducing magnetic interference fields.

Key point: Inductance

Inductance L describes a component that is capable of storing electrical energy in a magnetic field.

The calculation of the voltage of an inductor can be derived from the calculation of the voltage of an induction (equation 6.2).

$$u_{\mathrm{i}} = -N \cdot \frac{\mathrm{d}\Phi}{\mathrm{d}t} \tag{6.2}$$

The voltage of an inductance u is a change in current $\mathrm{d}i/\mathrm{d}t$ over time multiplied by the inductance L. Lenz's law also applies here, resulting in a negative sign.

$$u = -L \cdot \frac{\mathrm{d}i}{\mathrm{d}t} \tag{7.3}$$

In circuit diagrams, inductors are represented by black filled rectangles, as shown in the upper of the two figures 7.3. Less commonly, a coiled conductor is used to represent an inductor, as shown in the lower figure 7.3.

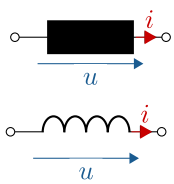


Figure 7.3: Circuit symbol for inductance

There are two ways to determine the inductance of a structure: by measuring the field strengths or by measuring the magnetic resistance.

Method 1 using field sizes:

1. Assumption of a Stromstärke $I \rightarrow$ flux
   $\Theta = N \cdot I$
2. Calculate field strength
   $H = \frac{\Theta}{\ell_\mathrm{m}}$
3. Calculate flux density
   $B = \mu_\mathrm{r} \cdot \mu_0 \cdot H$
4. Determine magnetic flux
   $\Phi = \iint_A \vec{B} \cdot \mathrm{d}\vec{A}$
5. Inductance
   $L = \frac{N \cdot \Phi}{I}$

Method 2 via magnetic resistance:

1. Divide the magnetic circuit into sections and magnetic resistance R_m calculate
   $R_\mathrm{m} = \frac{\ell_\mathrm{m}}{\mu_\mathrm{r} \cdot \mu_0 \cdot A}$
2. Calculate total resistance (for series structure)
   $R_{m,\mathrm{ges}} = \sum R_\mathrm{m}$
3. Inductance
   $L = \frac{N^2}{R_{m,\mathrm{ges}}}$

Example 7.1: Inductance

In a toroidal coil with an air gap, the influence of the air gap d on the inductance L is to be investigated. The coil has $N = 100$ turns. The relative permeability is $\mu_\mathrm{r} = 2000$. The average iron core length is $\ell_\mathrm{m} = 5\,\mathrm{cm}$. The cross-sectional area is $A = 1\,\mathrm{cm}^2$.

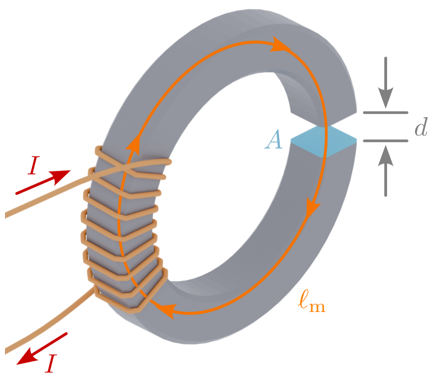


Figure 7.4: **Toroidal coil with air gap.**

- The air gap does not exist for the time being: $d = 0$. How large is the inductance L?

$$
\begin{aligned}
R_\mathrm{m} &= \frac{\ell_\mathrm{m}}{\mu_\mathrm{r} \cdot \mu_0 \cdot A} \\
L &= \frac{N^2}{R_m} \\
&= \frac{100^2 \cdot 2000 \cdot 1{,}256 \cdot 10^{-6}\,\frac{\mathrm{Vs}}{\mathrm{Am}} \cdot 1 \cdot 10^{-4}\,\mathrm{m}^2}{0{,}05\,\mathrm{m}} \\
L &= 0{,}05\,\frac{\mathrm{Vs}}{\mathrm{A}} = 50\,\mathrm{mH}
\end{aligned}
$$

- The air gap is now $d = 1\,\mathrm{mm}$. How large is the inductance?

$$
\begin{aligned}
R_\mathrm{m} &= R_{m,\mathrm{Fe}} + R_{m,\mathrm{L}} \\
L &= \frac{\ell_\mathrm{m} - d}{\mu_\mathrm{r} \cdot \mu_0 \cdot A} + \frac{d}{\mu_0 \cdot A} \\
L &= \frac{0{,}05\,\mathrm{m} - 0{,}001\,\mathrm{m}}{2000 \cdot 1{,}256 \cdot 10^{-6}\,\frac{\mathrm{Vs}}{\mathrm{Am}} \cdot 1 \cdot 10^{-4}\,\mathrm{m}^2} + \frac{0{,}001\,\mathrm{m}}{1{,}256 \cdot 10^{-6}\,\frac{\mathrm{Vs}}{\mathrm{Am}} \cdot 1 \cdot 10^{-4}\,\mathrm{m}^2} \\
L &= 1{,}226\,\mathrm{mH}
\end{aligned}
$$

8 Energy in the magnetic field

To calculate the energy stored in a coil, the toroidal coil illustrated in Figure 8.1 is connected to a DC voltage source. At time $t = 0$, the voltage source is switched on. The voltage across the coil is therefore equal to the source voltage U at every point in time $t > 0$. The current flowing through the coil will increase linearly according to equation 7.3. The energy supplied in this way generates a flux density in the coil core. To calculate the energy supplied to the coil, the known equation for electrical power 8.1 can be transformed and then the voltage can be replaced by equation 7.3. As a result of this transformation, the change in magnetic energy $\mathrm{d}W_\mathrm{L}$ is obtained, which is calculated from the multiplication of inductance L, the current through the inductance i_L and the rate of change of the current through the inductance $\mathrm{d}i_\mathrm{L}$ (see equation 8.2).

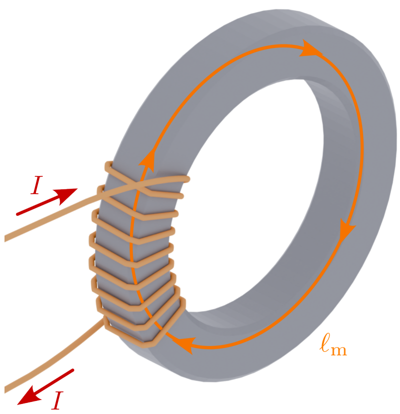


Figure 8.1: **Closed toroidal coil.**

$$p = \frac{\mathrm{d}W}{\mathrm{d}t} = u \cdot i \tag{8.1}$$

$$u = L \cdot \frac{\mathrm{d}i}{\mathrm{d}t} \tag{7.3}$$

$$\mathrm{d}W_\mathrm{m} = u_\mathrm{L} \cdot i_\mathrm{L} \cdot \mathrm{d}t = L \cdot \frac{\mathrm{d}i}{\mathrm{d}t} \cdot i_\mathrm{L} \cdot \mathrm{d}t = L \cdot i_\mathrm{L} \cdot \mathrm{d}i_\mathrm{L} \tag{8.2}$$

The total energy is the integral over $\mathrm{d}W_\mathrm{m}$ from the initial value $i_\mathrm{L} = 0$ to the static final value $i_\mathrm{L} = I$. Assuming a constant inductance L, the following applies:

$$W_m = L \cdot \int_0^{I_L} i_L \cdot \mathrm{d}i_L = \frac{1}{2} \cdot L \cdot I^2 = \frac{1}{2} \cdot N \cdot \Phi \cdot I \tag{8.3}$$

The magnetic energy can also be calculated from the field magnitudes by integration with the magnetic flux density:

$$W_\mathrm{m} = \ell_\mathrm{m} \cdot A \cdot \int_0^{B_L} H \mathrm{d}B = \ell_\mathrm{m} \cdot A \cdot \frac{B_\mathrm{L}^2}{2 \cdot \mu_\mathrm{r} \cdot \mu_0} \tag{8.4}$$

In an electromagnet, the energy density in the air gap (i.e. the energy per volume of the air gap) is particularly important: it represents the „force of the electromagnet“. The maximum force is exerted precisely at the transition from the air gap to the iron core – corresponding to a theoretically infinitely narrow air gap. Since the air gap, as the name suggests, is usually filled with air, $\mu_\mathrm{r} = 1$ in normal cases and can be omitted.

$$\frac{W_\mathrm{m}}{V} = \frac{B_\mathrm{L}^2}{2 \cdot \mu_0} = \frac{F}{A} \tag{8.5}$$

9 Skin-Effect

The skin effect describes the phenomenon whereby the current flow in a conductor carrying alternating current shifts from the core to its outer edge. This effect is based on the principle of induction and therefore only occurs in alternating current transmission.

Alternating current causes constantly changing magnetic fields. These magnetic fields counteract the cause according to Lenz's law and are more pronounced inside the conductor than at the edges. This results in a weakening of the electric field inside the conductor, so that the current is transmitted to the edge of the conductor. The extent to which the alternating current penetrates the conductor is described by the skin depth δ (delta). As the frequency increases, the field weakening in the core increases. Table 9.1 illustrates the effects of the skin effect using the example of a copper conductor. While the skin depth is 29.7 mm at a frequency of 5 Hz, it shrinks to 2.97 mm at a frequency of 500 Hz.

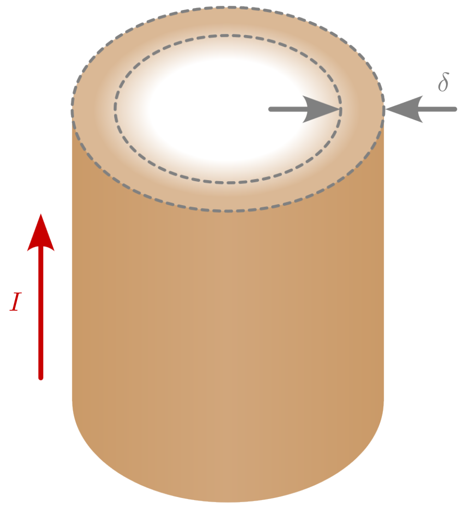


Figure 9.1: **Skin-Effect** The skin depth δ describes how deep the alternating current penetrates into the conductor.

Table 9.1: **Exemplary overview of skin depth.** Penetration depth into a copper cable depending on frequency:

Frequency	**Skin-Depth** δ_{Cu}
5 Hz	29,7 mm
50 Hz	9,38 mm
500 Hz	2,97 mm
500 kHz	93,8 μm

To calculate the skin effect, the skin depth δ is determined. The skin depth is calculated by dividing the square root of the specific resistance ρ_R by the product of the frequency f, the number pi π and the material-dependent magnetic permeability μ (see Equation 9.1).

$$\delta = \sqrt{\frac{\rho_\mathrm{R}}{\pi \cdot \mu \cdot f}} \tag{9.1}$$

Key point: Skin-Effekt

The skin effect causes alternating current to be displaced to the edge of the conductor.

Example 9.1: Skin-Depth

A copper conductor is traversed by a current with a frequency of $f = 50\,\mathrm{Hz}$. The following values are given for the conductor:

- Absolute magnetic permeability: $\mu = 4\pi \cdot 10^{-7}\,\frac{\mathrm{Vs}}{\mathrm{Am}}$
- Specific resistance: $\rho_R = 0{,}01721\,\frac{\Omega \cdot \mathrm{mm}^2}{\mathrm{m}}$

Calculate the skin depth δ of the conductor.

$$\delta = \sqrt{\frac{\rho_\mathrm{R}}{\pi \cdot \mu \cdot f}}$$
$$\delta = \sqrt{\frac{0{,}01721\,\frac{\Omega \cdot \mathrm{mm}^2}{\mathrm{m}}}{\pi \cdot 4\pi \cdot 10^{-7}\,\frac{\mathrm{Vs}}{\mathrm{Am}} \cdot 50\,\mathrm{Hz}}}$$
$$\delta \approx 9{,}34\,\mathrm{mm}$$

10 Hall-Effect

The Hall effect describes the occurrence of an electrical voltage within a current-carrying conductor due to the displacement of charge by a magnetic field.

In Figure 10.1, the conductor is traversed by a magnetic field, similar to the experimental setup for the Lorentz force. However, the conductor is shown here in plate form so that the charge carriers can move across the entire surface of the conductor. When current flows through the conductor, the Lorentz force acts on the moving charges. According to the right-hand rule, this causes a perpendicular deflection of the charge carriers in the conductor towards the edge. The charge displacement creates an electric field in the conductor that is opposite to the Lorentz force. The voltage difference resulting from the charge displacement can be measured at the edges of the conductor. It is referred to as the Hall voltage U_H.

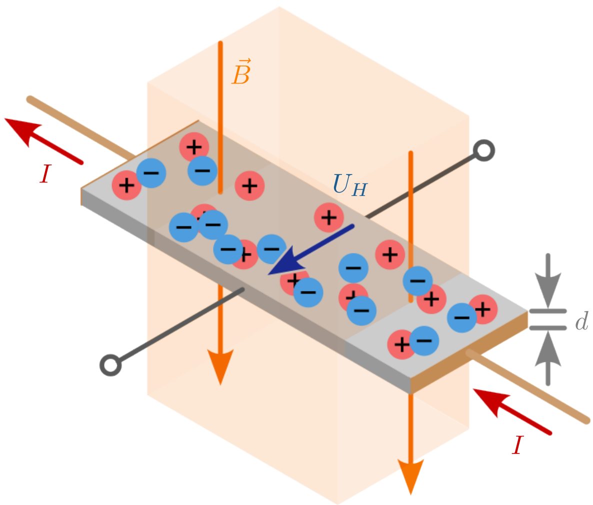


Figure 10.1: **Experimental setup for the Hall effect.** The Lorentz force generated by the magnetic field deflects the moving charge carriers perpendicularly to the edge. This charge displacement generates an electrically measurable voltage - the Hall voltage.

The Hall constant A_H is required to determine the Hall voltage U_H. The Hall constant A_H is a material parameter that provides information about the charge carrier density and thus about the conductivity of the material.

The Hall voltage can be calculated (as shown in equation 10.1) by multiplying the Hall constant A_H by the product of the current I and the magnetic flux density B divided by the thickness of the current-carrying plate d.

$$U_\mathrm{H} = A_\mathrm{H} \cdot \frac{I \cdot B}{d} \tag{10.1}$$

Key point: Hall-Effect

The Hall voltage U_H describes the occurrence of an electrical voltage within the current-carrying conductor perpendicular to the current flow due to an acting magnetic field.

Example 10.1: Hall-Voltage

A metal plate with a thickness of $d = 1\,\mathrm{cm}$ and a current of $I = 2\,\mathrm{A}$ is located in a magnetic field with a flux density of $B = 0.5\,\mathrm{T}$. The Hall constant of the material is $A_\mathrm{H} = 3.2 \cdot 10^{-3}\,\frac{\mathrm{m}^3}{\mathrm{C}}$. Calculate the Hall voltage.

$$U_\mathrm{H} = A_\mathrm{H} \cdot \frac{I \cdot B}{d}$$
$$U_\mathrm{H} = 3{,}2 \cdot 10^{-3}\,\frac{\mathrm{m}^3}{\mathrm{C}} \cdot \frac{2\,\mathrm{A} \cdot 0{,}5\,\mathrm{T}}{0{,}01\,\mathrm{m}}$$
$$U_\mathrm{H} = 0{,}32\,\mathrm{V}$$

A Exercise tasks

A.1 DC motor 1

An externally excited DC machine has the following information on its nameplate:

- Rated power: 41,8 kW
- Rated speed: 1900 $\frac{1}{\text{min}}$
- Anchor voltage: 440 V
- Anchor current: 100 A
- Excitement: 240 V
- Excitation current: 10 A
- Idling speed: 2000 $\frac{1}{\text{min}}$

Note: The rated power on a motor nameplate always refers to the mechanical power output at the rated point.

a) What is the torque of the machine at the rated point?

b) Calculate the efficiency of the machine at the rated point.

c) Calculate the product of the excitation flux and the armature constant $K \cdot \Phi$

d) How large is the anchor resistance R_A?

B Solutions to the exercises

B.1 DC motor 1

a) The rated torque is directly derived from the rated power and rated speed:

$$P_{\text{mech}} = M \cdot \omega$$
$$M = \frac{P_{\text{mech}}}{\omega} = \frac{41{,}8\,\text{kW}}{2\pi \cdot 1900\,\frac{1}{\text{min}} \cdot \frac{1\,\text{min}}{60\,\text{s}}} = 210{,}08\,\text{Nm}$$

b) In motor operation, the efficiency is the quotient of mechanical power relative to electrical power (see equation **??**). Both the armature power and the excitation power must be taken into account here.

$$\eta = \frac{P_{\text{mech}}}{P_{\text{elektr}}} = \frac{41{,}8\,\text{kW}}{440\,\text{V} \cdot 100\,\text{A} + 240\,\text{V} \cdot 10\,\text{A}} = 90{,}08\,\%$$

c) The induced voltage is calculated using equation **??**. When idling, the induced voltage must correspond to the armature voltage. The same can be seen from equation **??**, since the torque must also be zero when idling.

$$U_q = K \cdot \Phi \cdot \omega \qquad (??)$$
$$K \cdot \Phi = \frac{U_q}{\omega} = \frac{440\,\text{V}}{2\pi \cdot 2000\,\frac{1}{\text{min}} \cdot \frac{1\,\text{min}}{60\,\text{s}}} = 2{,}1\,\text{Vs}$$

d) There are two possible approaches here. Equation **??** can be transformed to the anchor resistance:

$$n = \frac{U_A}{2\pi K \cdot \Phi} - \frac{R_A \cdot M_i}{2\pi (K \cdot \Phi)^2} \qquad (\textbf{??})$$

$$R_A = \left(\frac{U_A}{2\pi K \cdot \Phi} - n \right) \cdot \frac{2\pi (K \cdot \Phi)^2}{M_i}$$

$$= \left(\frac{440\,\mathrm{V}}{2\pi \cdot 2{,}1\,\mathrm{Vs}} - \frac{1900}{60\,\mathrm{s}} \right) \cdot \frac{2\pi \cdot (2{,}1\,\mathrm{Vs})^2}{210{,}08\,\mathrm{Nm}} = 220\,\mathrm{m\Omega}$$

The other calculation variant uses the equivalent circuit diagram in Figure **??**. In nominal operation, we can calculate the source voltage U_q. This must be smaller than the source voltage in no-load operation.

$$U_q = K \cdot \Phi \cdot \omega \qquad (\textbf{??})$$

$$U_{q,n} = 2{,}1\,\mathrm{Vs} \cdot 2\pi \cdot 1900\,\frac{1}{\mathrm{min}} \cdot \frac{1\,\mathrm{min}}{60\,\mathrm{s}} = 418\,\mathrm{V}$$

The voltage drop across resistor R_A must be equal to the difference between the armature voltage and the induced voltage through the mesh circulation. Since the armature current is known in nominal operation, the resistance can be calculated using Ohm's law.

$$R_A = \frac{U_A - U_q}{I_A} = \frac{440\,\mathrm{V} - 418\,\mathrm{V}}{100\,\mathrm{A}} = 220\,\mathrm{m\Omega}$$

Index